

Section 2.1 - Game-Level Analysis: Baseline Ragression

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Description

This file is the third analysis file and the first of the game-level analysis (Section 2). This file only contains the baseline game-level regression model used in the thesis. This model takes all games that appear on both Steam and Epic and run a regression model to determine the effect of Epic entry. The model is run for four different pre/post windows (30, 60, 90, and 180 days). Although the specifics of the model changed frequently over the course of the project, the final model included time and game fixed effects.

Notes / Important Information

- Made the decision to drop Puzzle and Rougelike due to their low observation count (n=12 and n=15)
 - Dropped Time_Since_Release from the regression models due to the unlikelihood of depreciation occurring in the time window. Significance of post variables do not change.
-

Setup and Data Cleaning

```
# Libraries
library(tidyverse)
library(fixest)
library(broom)
library(ggplot2)
library(flextable)
library(modelsummary)
library(stargazer)

# Data
data <- read_csv("Data/cleaned_data.csv")
```

Find Epic Entry Date for each Game

```
epic_entry_dates <- data %>%
  filter(shop_name == "Epic") %>%
  group_by(game_title) %>%
  # entry date = first date the game appears on Epic
  summarise(entry_date = min(date), .groups = "drop")
```

Genre columns and window vector

```
genre_cols <- c("RPG", "Action", "Adventure", "Shooter", "Strategy", "Simulation",
               "Platformer", "Horror", "OpenWorld", "Survival",
               "Multiplayer", "Narrative")

# the four pre/post windows used throughout this file
windows <- c(30, 60, 90, 180)

# the widest window we ever need; build the panel once at this width and
# subset to narrower windows on demand (avoids rebuilding the panel 4x)
max_window <- max(windows)
```

Build Steam panel at the widest window

```
steam_panel <- data %>%
  filter(shop_name == "Steam") %>%
  left_join(epic_entry_dates, by = "game_title") %>%
  mutate(
    days_since_entry = as.numeric(date - entry_date),
    ever_treated     = !is.na(entry_date),
    post             = if_else(ever_treated, days_since_entry >= 0, FALSE)
  ) %>%
  filter(!ever_treated | abs(days_since_entry) <= max_window) %>%
  group_by(game_title, date) %>%
  summarise(
    price           = mean(price, na.rm = TRUE),
    discount_percent = mean(discount_percent, na.rm = TRUE),
    entry_date      = first(entry_date),
    release_date    = first(release_date),
    days_since_entry = first(days_since_entry),
    ever_treated    = first(ever_treated),
    post           = first(post),
    across(all_of(genre_cols), first),
    sale           = first(sale),
    .groups        = "drop"
  )
```

Clean up Loose Ends

Some games caused issues due to lack of price variation or other issues.

```
# Book of Demons: convert sub-$5 records to 4.99
steam_panel <- steam_panel %>%
  mutate(price = if_else(game_title == "Book of Demons" & price < 5, 4.99, price))

# drop problematic titles
steam_panel <- steam_panel %>%
  filter(!game_title %in% c("Dying Light", "Grounded", "The Sims 4"))
```

Add Analysis Variables (treated games only)

```

treated_panel <- steam_panel %>%
  filter(ever_treated) %>%
  mutate(
    on_epic          = as.integer(days_since_entry >= 0),
    time_since_release = as.numeric(date - release_date),
    period           = if_else(days_since_entry < 0, "pre", "post"),
    cal_week          = as.integer(floor(as.numeric(date - min(date)) / 7))
  )

cat("Treated games (180-day panel):", n_distinct(treated_panel$game_title), "\n")

## Treated games (180-day panel): 324

```

Regression Analysis

Similar to Section 1's regression model, I included different variations (no game FE) of the regression models in the code despite not using them in the thesis.

1. Helper: run base + genre regression for one outcome at one window

```

genre_inter <- paste0("on_epic:", genre_cols, collapse = " + ")

run_regs <- function(df, outcome_col, w) {
  d <- df %>% filter(abs(days_since_entry) <= w)

  base <- feols(
    as.formula(paste(outcome_col,
                      "~ on_epic | game_title + cal_week")),
    data = d, cluster = ~ game_title
  )

  genre <- feols(
    as.formula(paste(outcome_col,
                      "~ on_epic +", genre_inter,
                      "| game_title + cal_week")),
    data = d, cluster = ~ game_title
  )

  list(base = base, genre = genre)
}

```

2. Price regressions across windows

```

price_models <- map(windows, ~ run_regs(treated_panel, "price", .x))
names(price_models) <- paste0("w", windows)

# Base Model
etable(map(price_models, "base"),
  headers = paste0(windows, "d"),
  title = "Price ~ on_epic (base) across windows")

```

```
##                                w30                                w60                                w90
##                                30d                                60d                                90d
## Dependent Var.:              price                                price                                price
##
## on_epic                      -0.2361 (0.4475) -0.1826 (0.3725) 0.0392 (0.3410)
## Fixed-Effects: -----
## game_title                    Yes                                Yes                                Yes
## cal_week                      Yes                                Yes                                Yes
## -----
## S.E.: Clustered by: game_title by: game_title by: game_title
## Observations                  17,817                          34,933                          51,731
## R2                            0.86420                          0.84317                          0.83486
## Within R2                     9.39e-5                          5.03e-5                          2.26e-6
##
##                                w180
##                                180d
## Dependent Var.:              price
##
## on_epic                      0.1800 (0.2633)
## Fixed-Effects: -----
## game_title                    Yes
## cal_week                      Yes
## -----
## S.E.: Clustered by: game_title
## Observations                  101,065
## R2                            0.81979
## Within R2                     4.6e-5
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Genre Model
etable(map(price_models, "genre"),
       headers = paste0(windows, "d"),
       title = "Price ~ on_epic + genres across windows")
```

```
##                                w30                                w60                                w90
##                                30d                                60d                                90d
## Dependent Var.:              price                                price                                price
##
## on_epic                      -0.3303 (0.9111) -0.9108 (0.6780) -0.0126 (0.5845)
## on_epic x RPG                 0.3854 (0.8902)  1.048 (0.8560)  0.4713 (0.7014)
## on_epic x Action              0.2035 (0.7832)  0.3243 (0.5608)  0.3675 (0.4875)
## on_epic x Adventure           0.4315 (0.7688)  0.6211 (0.5137) -0.0391 (0.4602)
## on_epic x Shooter            -0.5198 (1.270)  0.6760 (0.8111)  0.6091 (0.6461)
## on_epic x Strategy            1.706 (1.040)   1.630 (1.114)   1.058 (0.8711)
## on_epic x Simulation          0.3267 (0.9442) -0.9716 (0.7288) -0.6336 (0.6518)
## on_epic x Platformer         0.0345 (1.065)  0.7790 (0.8218)  0.2582 (0.7516)
## on_epic x Horror             -0.3187 (1.234) -0.4121 (0.6943) -0.5932 (0.5358)
## on_epic x OpenWorld          -0.4680 (0.8368) -0.5075 (0.6657) -0.9249 (0.5768)
## on_epic x Survival            -0.4401 (0.9173)  1.107 (0.6992)  0.6852 (0.5759)
## on_epic x Multiplayer        -0.7480 (0.8141) -0.2434 (0.5501) -0.2819 (0.4708)
## on_epic x Narrative          -0.8369 (0.9336) -0.3204 (0.6329) -0.6629 (0.5375)
## Fixed-Effects: -----
## game_title                    Yes                                Yes                                Yes
## cal_week                      Yes                                Yes                                Yes
```

```
## -----
## S.E.: Clustered      by: game_title  by: game_title  by: game_title
## Observations          17,817          34,933          51,731
## R2                    0.86477          0.84375          0.83530
## Within R2             0.00425          0.00379          0.00268
##
##                      w180
##                      180d
## Dependent Var.:      price
##
## on_epic               0.0684 (0.4407)
## on_epic x RPG          1.184* (0.5344)
## on_epic x Action      -0.0498 (0.4661)
## on_epic x Adventure    0.1946 (0.3799)
## on_epic x Shooter      0.6227 (0.5459)
## on_epic x Strategy     0.0035 (0.6590)
## on_epic x Simulation   0.2797 (0.5250)
## on_epic x Platformer   0.2782 (0.5789)
## on_epic x Horror       0.1450 (0.5530)
## on_epic x OpenWorld   -0.5299 (0.4381)
## on_epic x Survival     0.3050 (0.5316)
## on_epic x Multiplayer -0.6286 (0.4638)
## on_epic x Narrative    -1.180* (0.4737)
## Fixed-Effects: -----
## game_title             Yes
## cal_week               Yes
## -----
## S.E.: Clustered      by: game_title
## Observations          101,065
## R2                    0.82020
## Within R2             0.00232
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3. Sale-frequency regressions across windows

```
sale_models <- map(windows, ~ run_regs(treated_panel, "sale", .x))
```

```
## NOTE: 1/1 fixed-effect singletons were removed (2 observations).
## NOTE: 1/1 fixed-effect singletons were removed (2 observations).
```

```
names(sale_models) <- paste0("w", windows)
```

```
etable(map(sale_models, "base"),
  headers = paste0(windows, "d"),
  title = "Sale ~ on_epic (base) across windows")
```

```
##                      w30          w60          w90
##                      30d          60d          90d
## Dependent Var.:      sale          sale          sale
##
## on_epic               0.0229 (0.0252) -0.0027 (0.0209) -0.0016 (0.0190)
## Fixed-Effects: -----
## game_title             Yes          Yes          Yes
## cal_week               Yes          Yes          Yes
```

```
## -----
## S.E.: Clustered by: game_title by: game_title by: game_title
## Observations 17,817 34,933 51,731
## R2 0.42159 0.32636 0.30033
## Within R2 0.00032 4.05e-6 1.41e-6
##
## w180
## 180d
## Dependent Var.: sale
##
## on_epic -0.0090 (0.0140)
## Fixed-Effects: -----
## game_title Yes
## cal_week Yes
## -----
## S.E.: Clustered by: game_title
## Observations 101,065
## R2 0.29546
## Within R2 4.36e-5
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
etable(map(sale_models, "genre"),
  headers = paste0(windows, "d"),
  title = "Sale ~ on_epic + genres across windows")
```

```
## w30 w60 w90
## 30d 60d 90d
## Dependent Var.: sale sale sale
##
## on_epic 0.0419 (0.0516) 0.0385 (0.0412) 0.0088 (0.0335)
## on_epic x RPG 0.0369 (0.0474) 0.0135 (0.0392) 0.0258 (0.0340)
## on_epic x Action -0.0256 (0.0427) -0.0093 (0.0317) -0.0122 (0.0256)
## on_epic x Adventure -0.0405 (0.0468) -0.0416 (0.0366) -0.0136 (0.0322)
## on_epic x Shooter 0.1361. (0.0739) 0.0519 (0.0551) 0.0325 (0.0448)
## on_epic x Strategy -0.1286* (0.0504) -0.1155** (0.0400) -0.0857* (0.0347)
## on_epic x Simulation -0.0097 (0.0549) -0.0164 (0.0423) -0.0299 (0.0377)
## on_epic x Platformer 0.0399 (0.0612) -0.0388 (0.0483) -0.0174 (0.0409)
## on_epic x Horror 0.0035 (0.0679) -0.0187 (0.0443) -0.0143 (0.0396)
## on_epic x OpenWorld -0.0369 (0.0343) -0.0086 (0.0290) 0.0255 (0.0265)
## on_epic x Survival 0.1126* (0.0493) 0.0499 (0.0489) 0.0658 (0.0446)
## on_epic x Multiplayer -0.0143 (0.0493) -0.0139 (0.0367) -0.0114 (0.0338)
## on_epic x Narrative -0.0550 (0.0525) -0.0495 (0.0375) -0.0152 (0.0330)
## Fixed-Effects: -----
## game_title Yes Yes Yes
## cal_week Yes Yes Yes
## -----
## S.E.: Clustered by: game_title by: game_title by: game_title
## Observations 17,817 34,933 51,731
## R2 0.42781 0.32953 0.30254
## Within R2 0.01106 0.00470 0.00316
##
## w180
## 180d
## Dependent Var.: sale
```

```
##
## on_epic          0.0017 (0.0236)
## on_epic x RPG    -0.0026 (0.0213)
## on_epic x Action -0.0021 (0.0184)
## on_epic x Adventure -0.0183 (0.0242)
## on_epic x Shooter  0.0156 (0.0323)
## on_epic x Strategy -0.0553* (0.0263)
## on_epic x Simulation -0.0289 (0.0309)
## on_epic x Platformer 0.0118 (0.0249)
## on_epic x Horror   -0.0483 (0.0380)
## on_epic x OpenWorld 0.0084 (0.0181)
## on_epic x Survival  0.0580. (0.0312)
## on_epic x Multiplayer -0.0042 (0.0280)
## on_epic x Narrative 0.0132 (0.0261)
## Fixed-Effects: -----
## game_title          Yes
## cal_week            Yes
## -----
## S.E.: Clustered      by: game_title
## Observations         101,065
## R2                   0.29654
## Within R2            0.00157
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4. Discount-depth regressions across windows

```
disc_models <- map(windows, ~ run_regs(treated_panel, "discount_percent", .x))
```

```
## NOTE: 1/1 fixed-effect singletons were removed (2 observations).
```

```
## NOTE: 1/1 fixed-effect singletons were removed (2 observations).
```

```
names(disc_models) <- paste0("w", windows)
```

```
etable(map(disc_models, "base"),
  headers = paste0(windows, "d"),
  title = "Discount % ~ on_epic (base) across windows")
```

```
##                               w30                w60                w90
##                               30d                60d                90d
## Dependent Var.: discount_percent discount_percent discount_percent
##
## on_epic          0.2007 (1.406)  0.1368 (1.118) -0.1852 (0.9958)
## Fixed-Effects: -----
## game_title          Yes                Yes                Yes
## cal_week            Yes                Yes                Yes
## -----
## S.E.: Clustered      by: game_title  by: game_title  by: game_title
## Observations         17,817         34,933         51,731
## R2                   0.41583         0.32611         0.30945
## Within R2            6.9e-6         2.88e-6         5.3e-6
##
##                               w180
##                               180d
## Dependent Var.: discount_percent
```

```
##
## on_epic          -0.1572 (0.7185)
## Fixed-Effects:  -----
## game_title      Yes
## cal_week        Yes
## -----
## S.E.: Clustered by: game_title
## Observations    101,065
## R2              0.29562
## Within R2       3.83e-6
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
etable(map(disc_models, "genre"),
        headers = paste0(windows, "d"),
        title = "Discount % ~ on_epic + genres across windows")
```

```
##                                w30          w60          w90
##                                30d          60d          90d
## Dependent Var.: discount_percent discount_percent discount_percent
##
## on_epic          1.890 (2.861)    2.985 (2.009)    0.6005 (1.606)
## on_epic x RPG    0.9333 (2.425)   -0.2477 (1.805)  0.8991 (1.528)
## on_epic x Action -2.021 (2.265)   -1.218 (1.618)  -1.676 (1.306)
## on_epic x Adventure -1.219 (2.392)   -1.398 (1.641)  0.5697 (1.349)
## on_epic x Shooter  2.715 (4.024)   -0.5498 (2.735)  0.0573 (1.993)
## on_epic x Strategy -7.995** (2.858) -6.876*** (1.944) -4.992** (1.523)
## on_epic x Simulation -0.9353 (2.865)    1.572 (1.765)  0.9650 (1.519)
## on_epic x Platformer 0.2766 (3.331)   -4.364. (2.552) -3.088 (2.187)
## on_epic x Horror    0.7910 (3.662)    0.0995 (2.169)  1.150 (1.720)
## on_epic x OpenWorld -0.9400 (1.963)   -0.1540 (1.470)  1.531 (1.352)
## on_epic x Survival  3.050 (2.483)   -0.9123 (2.116)  0.2784 (1.762)
## on_epic x Multiplayer -0.2798 (2.501)  -0.5051 (1.640) -0.4056 (1.402)
## on_epic x Narrative -0.4959 (2.586)   -1.609 (1.579)  -0.2801 (1.344)
## Fixed-Effects:  -----
## game_title      Yes              Yes              Yes
## cal_week        Yes              Yes              Yes
## -----
## S.E.: Clustered by: game_title by: game_title by: game_title
## Observations    17,817          34,933          51,731
## R2              0.41884          0.32829          0.31130
## Within R2       0.00517          0.00324          0.00268
##
##                                w180
##                                180d
## Dependent Var.: discount_percent
##
## on_epic          0.9592 (1.150)
## on_epic x RPG    -1.415 (0.9970)
## on_epic x Action -1.167 (0.9514)
## on_epic x Adventure -0.2702 (0.9638)
## on_epic x Shooter  0.0692 (1.284)
## on_epic x Strategy -2.915** (1.028)
## on_epic x Simulation -0.4464 (1.104)
## on_epic x Platformer -0.6150 (1.237)
```

```
## on_epic x Horror      -0.3000 (1.233)
## on_epic x OpenWorld   0.2974 (1.083)
## on_epic x Survival     0.8433 (1.352)
## on_epic x Multiplayer -0.2729 (1.083)
## on_epic x Narrative    0.8863 (1.023)
## Fixed-Effects: -----
## game_title            Yes
## cal_week              Yes
## -----
## S.E.: Clustered      by: game_title
## Observations         101,065
## R2                    0.29621
## Within R2            0.00084
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5. Persistence figure: on_epic coefficient vs window length

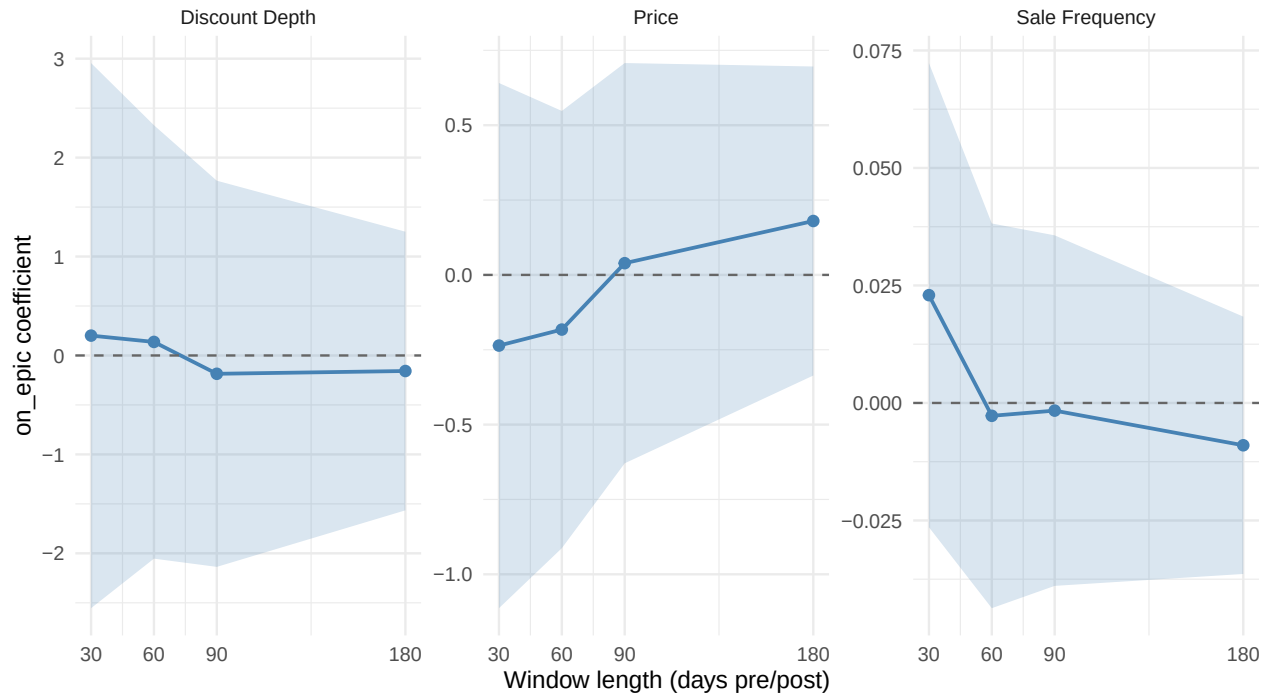
```
collect_on_epic <- function(models, label) {
  map_dfr(names(models), function(nm) {
    m <- models[[nm]]$base
    ct <- tidy(m) %>% filter(term == "on_epic")
    tibble(
      outcome = label,
      window  = as.integer(str_remove(nm, "w")),
      estimate = ct$estimate,
      se       = ct$std.error
    )
  })
}

persistence <- bind_rows(
  collect_on_epic(price_models, "Price"),
  collect_on_epic(sale_models,  "Sale Frequency"),
  collect_on_epic(disc_models,  "Discount Depth")
)

ggplot(persistence, aes(x = window, y = estimate)) +
  geom_ribbon(aes(ymin = estimate - 1.96 * se,
                 ymax = estimate + 1.96 * se),
            alpha = 0.2, fill = "steelblue") +
  geom_line(color = "steelblue", linewidth = 0.8) +
  geom_point(size = 2, color = "steelblue") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray40") +
  facet_wrap(~ outcome, scales = "free_y") +
  scale_x_continuous(breaks = windows) +
  labs(
    title   = "Epic-Entry Effect by Window Length (base specifications)",
    subtitle = "on_epic coefficient with 95% CI | treated games only",
    x       = "Window length (days pre/post)",
    y       = "on_epic coefficient"
  ) +
  theme_minimal()
```

Epic-Entry Effect by Window Length (base specifications)

on_epic coefficient with 95% CI | treated games only



Clean Regression Output

Like Section 1, I needed to make sure that my regression output could be used in actual thesis. These are just the Word outputs for the regression models.

```
genre_price <- map(price_models, "genre")
genre_sale <- map(sale_models, "genre")
genre_disc <- map(disc_models, "genre")
names(genre_price) <- names(genre_sale) <- names(genre_disc) <-
  paste0(windows, "d")

make_genre_table <- function(mods, caption) {
  modelsummary(
    mods,
    coef_map = c("on_epic" = "On Epic (post-entry)"),
    gof_omit = "AIC|BIC|Log.Lik.|RMSE|Std.Errors",
    stars    = c('*' = .05, '**' = .01, '***' = .001),
    title    = caption,
    notes    = paste("Genre x on_epic interactions estimated but omitted",
                     "from display. FE: game + calendar week.",
                     "SEs clustered by game."),
    output   = "flextable"
  ) %>%
  flextable::autofit()
}

price_genre_ft <- make_genre_table(genre_price,
```

```

"Price ~ on_epic (genre-interacted spec; genres omitted)"
sale_genre_ft <- make_genre_table(genre_sale,
  "Sale frequency ~ on_epic (genre-interacted spec; genres omitted)"
disc_genre_ft <- make_genre_table(genre_disc,
  "Discount depth ~ on_epic (genre-interacted spec; genres omitted)"

```

```
price_genre_ft
```

```

## Warning: fonts used in `flectable` are ignored because the `pdflatex` engine is
## used and not `xelatex` or `lualatex`. You can avoid this warning by using
## `set_flectable_defaults(fonts_ignore=TRUE)` or use a compatible engine (e.g.
## `pdf-engine: xelatex`).

```

Table 1: Price ~ on_epic (genre-interacted spec; genres omitted)

	30d	60d	90d	180d
On Epic (post-entry)	-0.330 (0.911)	-0.911 (0.678)	-0.013 (0.585)	0.068 (0.441)
Num.Obs.	17817	34933	51731	101065
R2	0.865	0.844	0.835	0.820
R2 Adj.	0.860	0.841	0.833	0.819
R2 Within	0.004	0.004	0.003	0.002
R2 Within Adj.	0.003	0.003	0.002	0.002
FE: game_title	X	X	X	X
FE: cal_week	X	X	X	X

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Genre x on_epic interactions estimated but omitted from display. FE: game + calendar week. SEs clustered by game.

```
sale_genre_ft
```

```

## Warning: fonts used in `flectable` are ignored because the `pdflatex` engine is
## used and not `xelatex` or `lualatex`. You can avoid this warning by using
## `set_flectable_defaults(fonts_ignore=TRUE)` or use a compatible engine (e.g.
## `pdf-engine: xelatex`).

```

Table 2: Sale frequency ~ on_epic (genre-interacted spec; genres omitted)

	30d	60d	90d	180d
On Epic (post-entry)	0.042 (0.052)	0.038 (0.041)	0.009 (0.033)	0.002 (0.024)
Num.Obs.	17817	34933	51731	101065
R2	0.428	0.330	0.303	0.297

Table 2: Sale frequency ~ on_epic (genre-interacted spec; genres omitted)

	30d	60d	90d	180d
R2 Adj.	0.406	0.317	0.294	0.292
R2 Within	0.011	0.005	0.003	0.002
R2 Within Adj.	0.010	0.004	0.003	0.001
FE: game_title	X	X	X	X
FE: cal_week	X	X	X	X

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Genre x on_epic interactions estimated but omitted from display. FE: game + calendar week. SEs clustered by game.

disc_genre_ft

```
## Warning: fonts used in `flectable` are ignored because the `pdflatex` engine is
## used and not `xelatex` or `lualatex`. You can avoid this warning by using
## `set_flectable_defaults(fonts_ignore=TRUE)` or use a compatible engine (e.g.
## `pdf-engine: xelatex`).
```

Table 3: Discount depth ~ on_epic (genre-interacted spec; genres omitted)

	30d	60d	90d	180d
On Epic (post-entry)	1.890	2.985	0.600	0.959
	(2.861)	(2.009)	(1.606)	(1.150)
Num.Obs.	17817	34933	51731	101065
R2	0.419	0.328	0.311	0.296
R2 Adj.	0.397	0.315	0.302	0.291
R2 Within	0.005	0.003	0.003	0.001
R2 Within Adj.	0.004	0.003	0.002	0.001
FE: game_title	X	X	X	X
FE: cal_week	X	X	X	X

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Genre x on_epic interactions estimated but omitted from display. FE: game + calendar week. SEs clustered by game.

```
save_as_docx(price_genre_ft, sale_genre_ft, disc_genre_ft,
             path = "Output/tab_genre_regs.docx")
```

```
base_price <- map(price_models, "base")
base_sale <- map(sale_models, "base")
base_disc <- map(disc_models, "base")
```

```

# window labels straight off the model lists (length-safe): "30d","60d",...
lab <- paste0(sub("^w", "", names(price_models)), "d")
names(base_price) <- names(base_sale) <- names(base_disc) <- lab

make_base_table <- function(mods, caption) {
  modelsummary(
    mods,
    coef_map = c("on_epic" = "On Epic (post-entry)"),
    gof_omit = "~R2$|R2 Adj\\.|R2 Within Adj\\.|AIC|BIC|Log\\.Lik\\.|RMSE|Std\\.Errors|FE:",
    stars    = c('*' = .05, '**' = .01, '***' = .001),
    title    = caption,
    output   = "flextable"
  ) %>%
  flextable::autofit()
}

price_base_ft <- make_base_table(base_price, "Price ~ On_Epic")
sale_base_ft  <- make_base_table(base_sale,  "Sale Frequency ~ On_Epic")
disc_base_ft  <- make_base_table(base_disc,  "Discount Depth ~ On_Epic")

price_base_ft; sale_base_ft; disc_base_ft

## Warning: fonts used in `flextable` are ignored because the `pdflatex` engine is
## used and not `xelatex` or `lualatex`. You can avoid this warning by using
## `set_flextable_defaults(fonts_ignore=TRUE)` or use a compatible engine (e.g.
## `pdf-engine: xelatex`).

```

Table 4: Price ~ On_Epic

	30d	60d	90d	180d
On Epic (post-entry)	-0.236 (0.447)	-0.183 (0.373)	0.039 (0.341)	0.180 (0.263)
Num.Obs.	17817	34933	51731	101065
R2 Within	0.000	0.000	0.000	0.000

* p < 0.05, ** p < 0.01, *** p < 0.001

```

## Warning: fonts used in `flextable` are ignored because the `pdflatex` engine is
## used and not `xelatex` or `lualatex`. You can avoid this warning by using
## `set_flextable_defaults(fonts_ignore=TRUE)` or use a compatible engine (e.g.
## `pdf-engine: xelatex`).

```

Table 5: Sale Frequency ~ On_Epic

	30d	60d	90d	180d
On Epic (post-entry)	0.023 (0.025)	-0.003 (0.021)	-0.002 (0.019)	-0.009 (0.014)
Num.Obs.	17817	34933	51731	101065

Table 5: Sale Frequency ~ On_Epic

	30d	60d	90d	180d
R2 Within	0.000	0.000	0.000	0.000

* p < 0.05, ** p < 0.01, *** p < 0.001

```
## Warning: fonts used in `flextable` are ignored because the `pdflatex` engine is
## used and not `xelatex` or `lualatex`. You can avoid this warning by using
## `set_flextable_defaults(fonts_ignore=TRUE)` or use a compatible engine (e.g.
## `pdf-engine: xelatex`).
```

Table 6: Discount Depth ~ On_Epic

	30d	60d	90d	180d
On Epic (post-entry)	0.201	0.137	-0.185	-0.157
	(1.406)	(1.118)	(0.996)	(0.718)
Num.Obs.	17817	34933	51731	101065
R2 Within	0.000	0.000	0.000	0.000

* p < 0.05, ** p < 0.01, *** p < 0.001

```
save_as_docx(price_base_ft, sale_base_ft, disc_base_ft,
             path = "Output/tab_base_regs.docx")
```

Extra Information and Code

Sample Diagnostics

```
steam_panel %>%
  filter(ever_treated) %>%           # or remove this to check all games
  distinct(game_title, .keep_all = TRUE) %>%
  summarise(across(all_of(genre_cols), sum, na.rm = TRUE)) %>%
  pivot_longer(everything(), names_to = "genre", values_to = "n_games") %>%
  arrange(desc(n_games))
```

```
## Warning: There was 1 warning in `summarise()`.
## i In argument: `across(all_of(genre_cols), sum, na.rm = TRUE)`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
## # Previously
## across(a:b, mean, na.rm = TRUE)
##
## # Now
## across(a:b, \(x) mean(x, na.rm = TRUE))
```

```
## # A tibble: 12 x 2
##   genre      n_games
##   <chr>      <dbl>
## 1 Adventure    98
## 2 Action      97
## 3 RPG         82
## 4 OpenWorld   71
## 5 Narrative    67
## 6 Multiplayer  56
## 7 Strategy    49
## 8 Simulation   45
## 9 Survival     45
## 10 Platformer  39
## 11 Horror      32
## 12 Shooter     31
```

```
steam_panel %>%
  distinct(game_title, ever_treated, .keep_all = TRUE) %>%
  group_by(ever_treated) %>%
  summarise(across(all_of(genre_cols), sum, na.rm = TRUE)) %>%
  pivot_longer(-ever_treated, names_to = "genre", values_to = "n_games") %>%
  pivot_wider(names_from = ever_treated, values_from = n_games,
              names_prefix = "treated_") %>%
  mutate(ratio = round(treated_FALSE / treated_TRUE, 1)) %>%
  arrange(ratio)
```

```
## # A tibble: 12 x 4
##   genre      treated_FALSE treated_TRUE ratio
##   <chr>      <dbl>      <dbl> <dbl>
## 1 Adventure    116         98  1.2
## 2 Platformer    52         39  1.3
## 3 OpenWorld     91         71  1.3
## 4 Survival      59         45  1.3
## 5 Narrative     84         67  1.3
## 6 Strategy      70         49  1.4
## 7 Horror        44         32  1.4
## 8 Simulation    70         45  1.6
## 9 Action       175         97  1.8
## 10 Shooter      60         31  1.9
## 11 Multiplayer  109         56  1.9
## 12 RPG         165         82  2
```

T-Test Analysis

Originally, I had also included a t-test analysis of the outcome variables that compared the Pre-Epic Steam data to the Post-Epic Steam data for each outcome variable. To display these results, I created graphs that plotted the values of each t-test in ascending order.

1. Helper: run per-game t-tests for one outcome at one window

```
# outcome_col: name of the column to test ("price", "sale", "discount_percent")
# w:          window length in days
run_ttests <- function(df, outcome_col, w, label) {
  df %>%
    filter(abs(days_since_entry) <= w) %>%
```

```

group_by(game_title) %>%
filter(sum(period == "pre") >= 5, sum(period == "post") >= 5) %>%
group_modify(~ {
  pre <- .x[[outcome_col]][.x$period == "pre"]
  post <- .x[[outcome_col]][.x$period == "post"]
  if (sd(pre) == 0 && sd(post) == 0) return(tibble())
  tidy(t.test(post, pre, alternative = "two.sided"))
}) %>%
ungroup() %>%
mutate(
  outcome      = label,
  window       = w,
  significant   = p.value < 0.05,
  direction_up = estimate > 0
)
}

```

2. Run all outcomes across all windows

```

outcome_specs <- tibble(
  col   = c("price", "sale", "discount_percent"),
  label = c("Price", "Sale Frequency", "Discount Depth")
)

# run t-tests (flat games excluded inside helper)
ttest_results <- map_dfr(windows, function(w) {
  map_dfr(seq_len(nrow(outcome_specs)), function(j) {
    run_ttests(treated_panel, outcome_specs$col[j], w, outcome_specs$label[j])
  })
}) %>%
mutate(no_change = FALSE)

# identify flat games for every outcome + window combination
flat_games <- map_dfr(windows, function(w) {
  map_dfr(seq_len(nrow(outcome_specs)), function(j) {
    col   <- outcome_specs$col[j]
    label <- outcome_specs$label[j]

    treated_panel %>%
      filter(abs(days_since_entry) <= w) %>%
      group_by(game_title) %>%
      filter(sum(period == "pre") >= 5, sum(period == "post") >= 5) %>%
      summarise(
        pre_sd = sd(.data[[col]][period == "pre"], na.rm = TRUE),
        post_sd = sd(.data[[col]][period == "post"], na.rm = TRUE),
        .groups = "drop"
      ) %>%
      filter(pre_sd == 0 & post_sd == 0) %>%
      mutate(
        estimate      = 0,
        p.value       = 1,
        significant    = FALSE,
        direction_up   = FALSE,

```

```

      no_change    = TRUE,
      outcome      = label,
      window       = w
    ) %>%
    select(game_title, estimate, p.value, significant,
           direction_up, no_change, outcome, window)
  })
})

# bind together
ttest_results <- bind_rows(ttest_results, flat_games)

# sanity check
ttest_results %>%
  count(outcome, window, no_change) %>%
  pivot_wider(names_from = no_change, values_from = n,
              names_prefix = "no_change_") %>%
  arrange(outcome, window)

```

```
## # A tibble: 12 x 4
##   outcome      window no_change_FALSE no_change_TRUE
##   <chr>         <dbl>         <int>         <int>
## 1 Discount Depth    30             208            59
## 2 Discount Depth    60             242            25
## 3 Discount Depth    90             250            17
## 4 Discount Depth   180             262             5
## 5 Price             30             208            59
## 6 Price             60             242            25
## 7 Price             90             250            17
## 8 Price            180             263             4
## 9 Sale Frequency    30             208            59
## 10 Sale Frequency   60             242            25
## 11 Sale Frequency   90             250            17
## 12 Sale Frequency  180             262             5

```

3. Plot helper: one t-test scatter for a given outcome + window

```

plot_ttests <- function(res, outcome_label, w, y_lab) {
  d <- res %>%
    filter(outcome == outcome_label, window == w) %>%
    arrange(estimate) %>%
    mutate(
      game_index = row_number(),
      color_group = case_when(
        no_change ~ "No change",
        direction_up & significant ~ "Rose (p<0.05)",
        !direction_up & significant ~ "Fell (p<0.05)",
        direction_up & !significant ~ "Rose (n.s.)",
        !direction_up & !significant ~ "Fell (n.s.)"
      )
    )
  ggplot(d, aes(x = game_index, y = estimate)) +

```

```

geom_point(aes(color = color_group), size = 2) +
geom_hline(yintercept = 0, linetype = "dashed", color = "gray40") +
scale_color_manual(
  values = c(
    "No change"      = "gray60",
    "Fell (n.s.)"    = "#93c4df",
    "Rose (n.s.)"    = "#f4a582",
    "Fell (p<0.05)"  = "steelblue",
    "Rose (p<0.05)"  = "firebrick"
  ),
  name = NULL,
  drop = FALSE
) +
labs(
  x      = "Games (sorted by change)",
  y      = y_lab,
  title  = paste0(outcome_label, " Change Within ", w,
                  " Days of Epic Entry"),
  subtitle = paste0("Each point = one game | n = ", nrow(d), " games")
) +
theme_minimal() +
theme(legend.position = "bottom")
}

```

4. Price t-test plots (one per window)

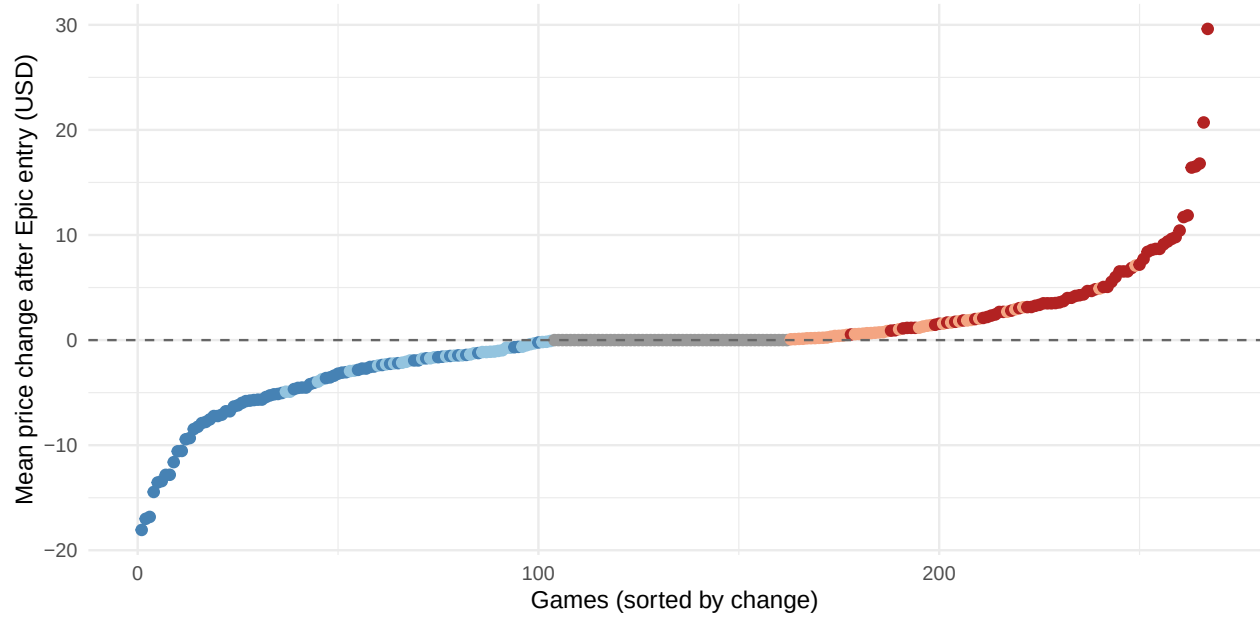
```

for (w in windows) {
  print(plot_ttests(ttest_results, "Price", w,
                    "Mean price change after Epic entry (USD)"))
}

```

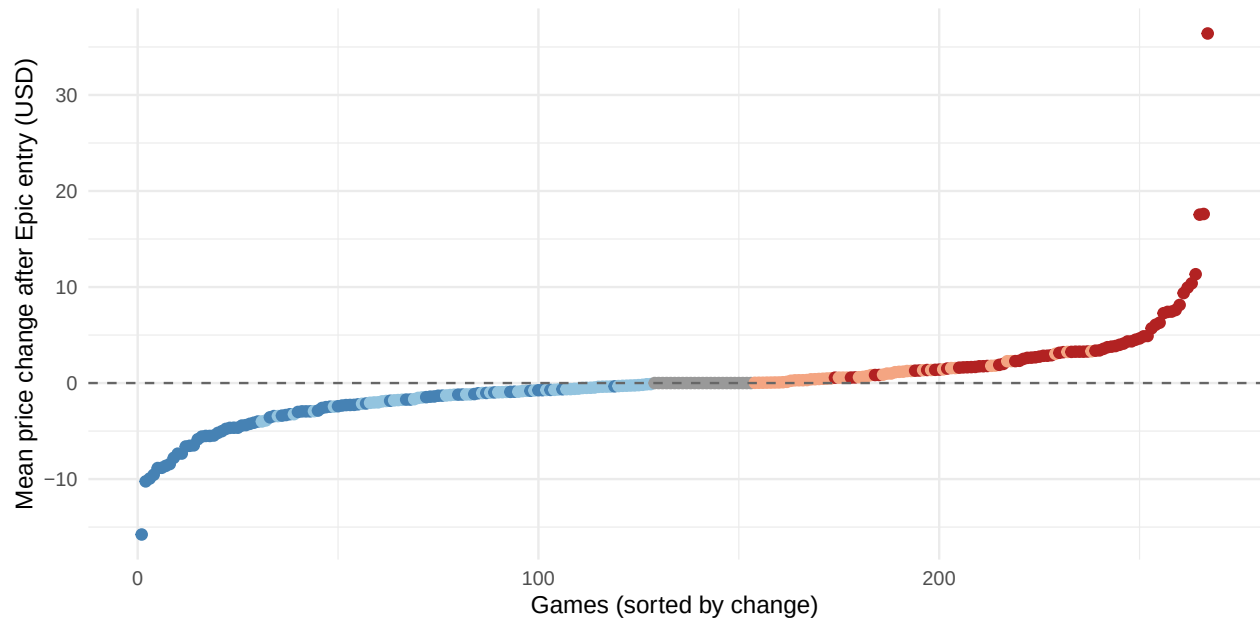
Price Change Within 30 Days of Epic Entry

Each point = one game | n = 267 games



Price Change Within 60 Days of Epic Entry

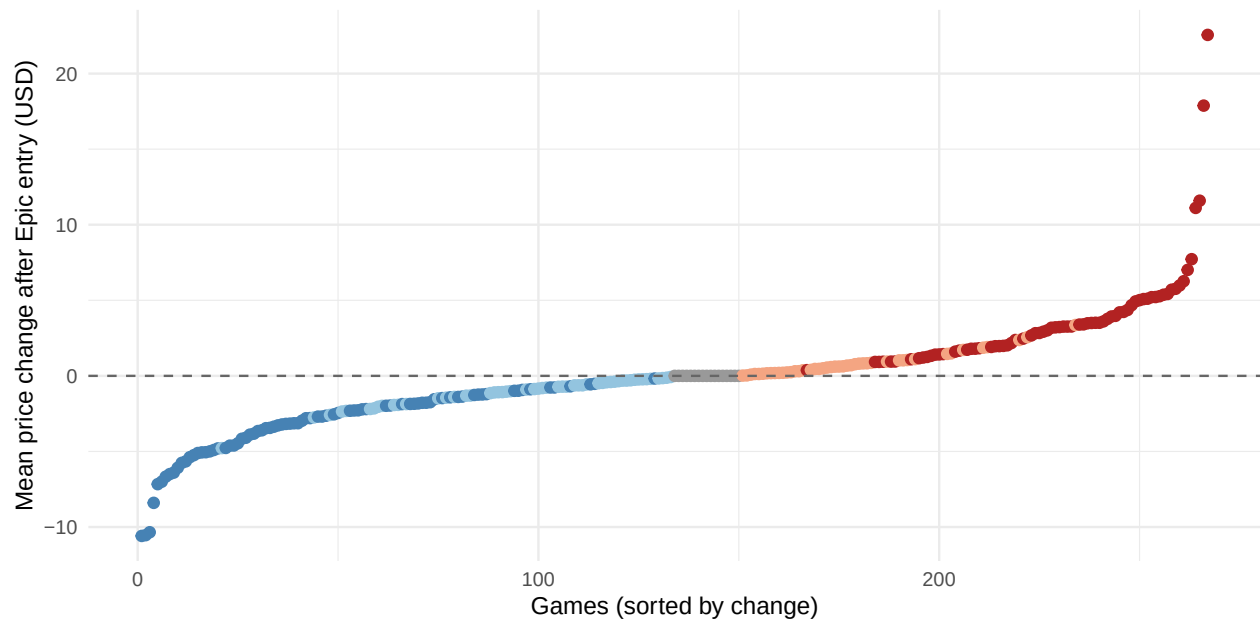
Each point = one game | n = 267 games



● Fell (n.s.) ● Fell (p<0.05) ● No change ● Rose (n.s.) ● Rose (p<0.05)

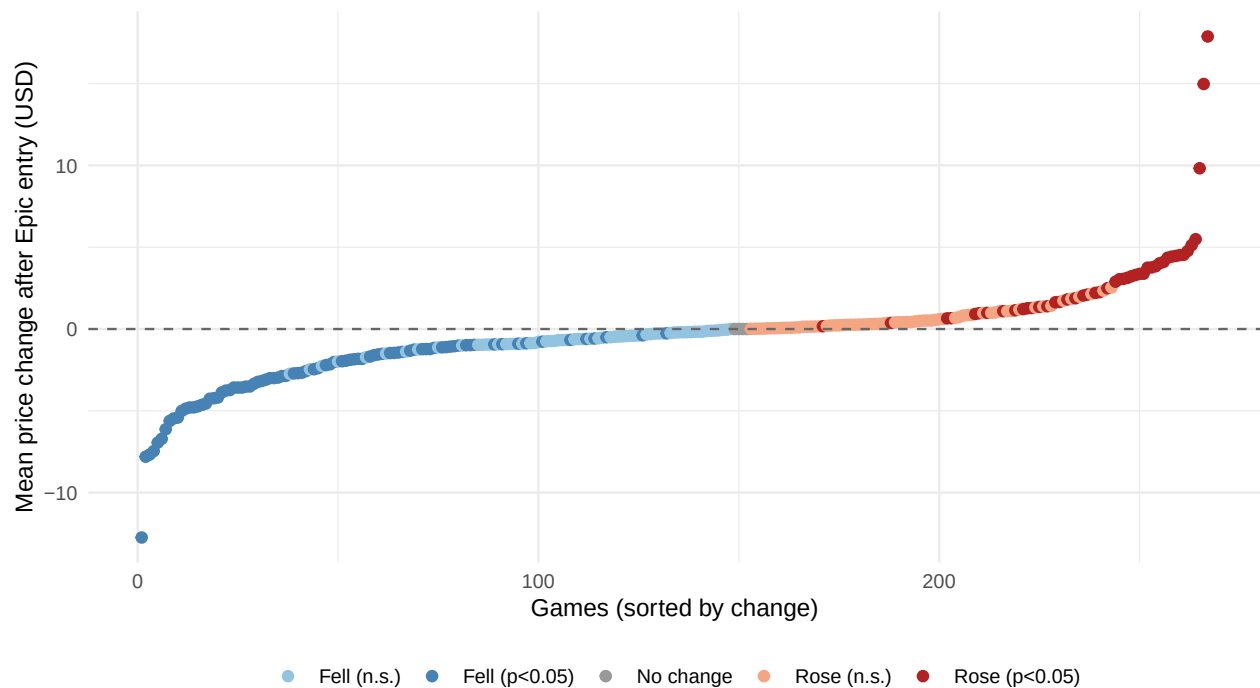
Price Change Within 90 Days of Epic Entry

Each point = one game | n = 267 games



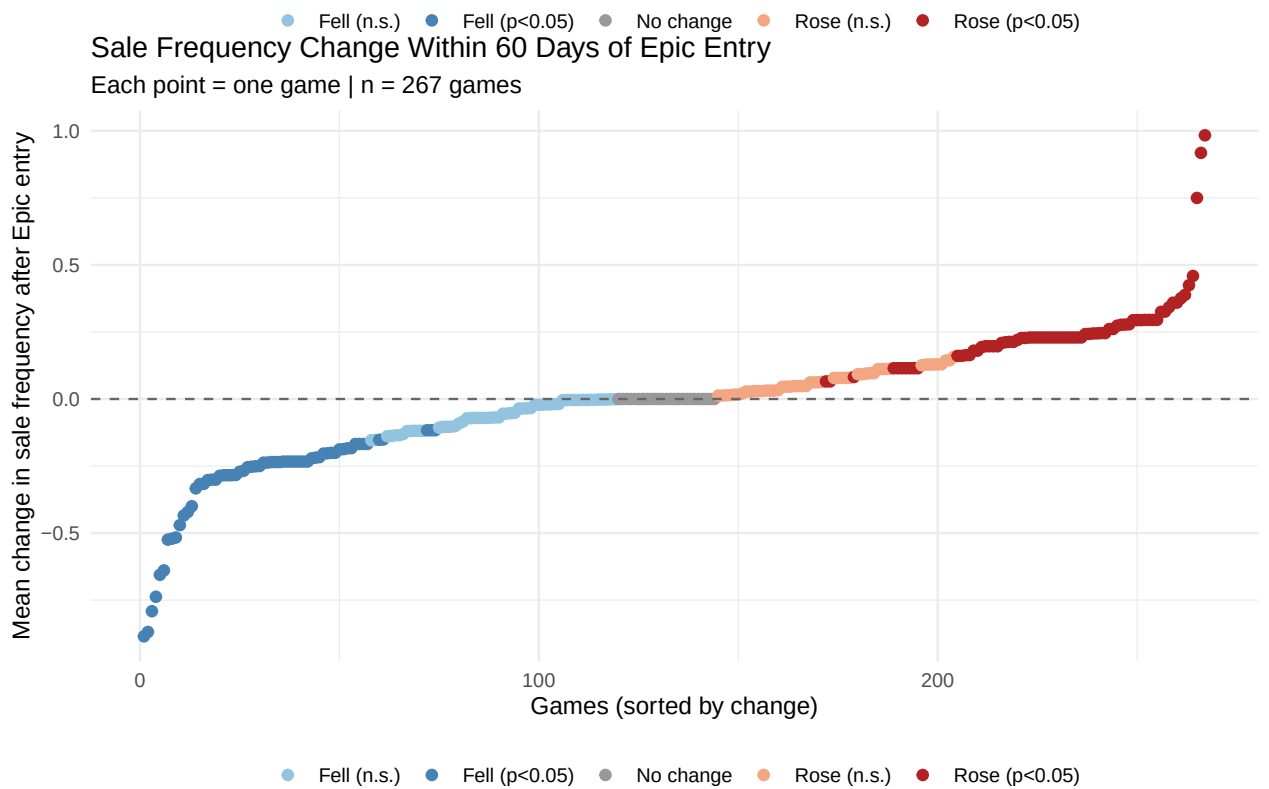
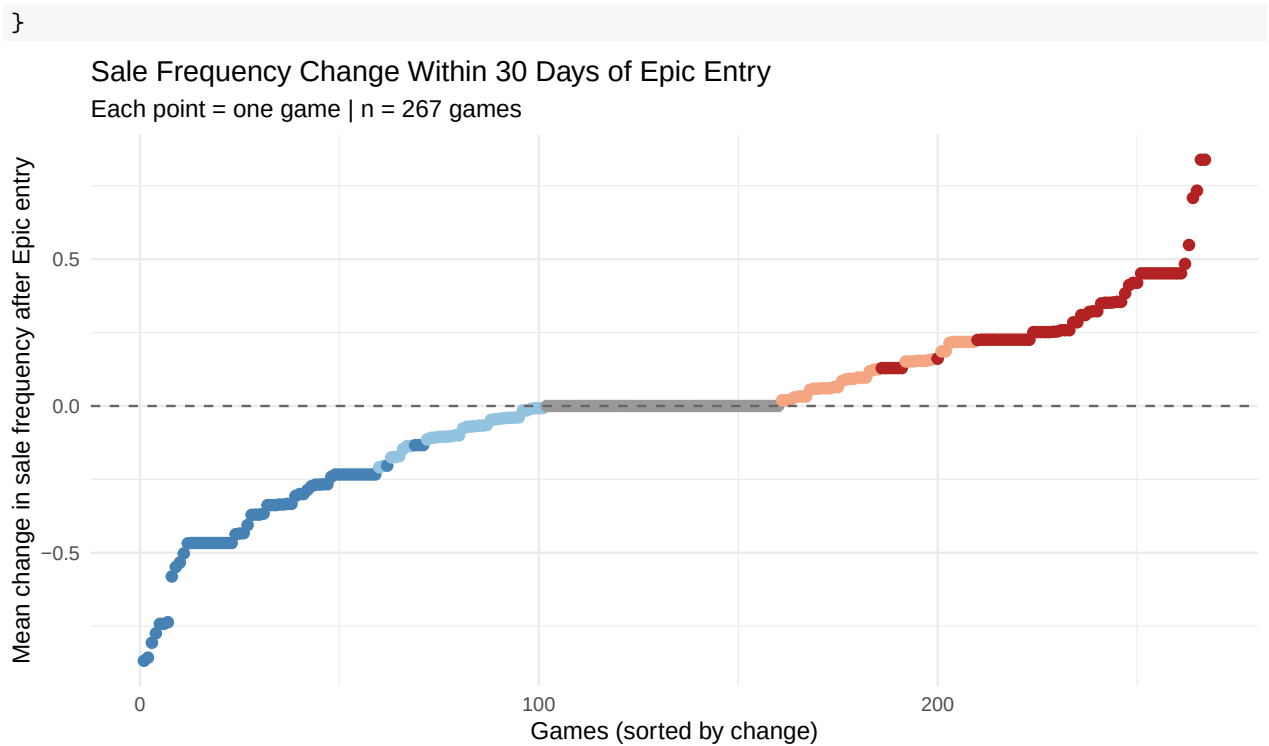
Price Change Within 180 Days of Epic Entry

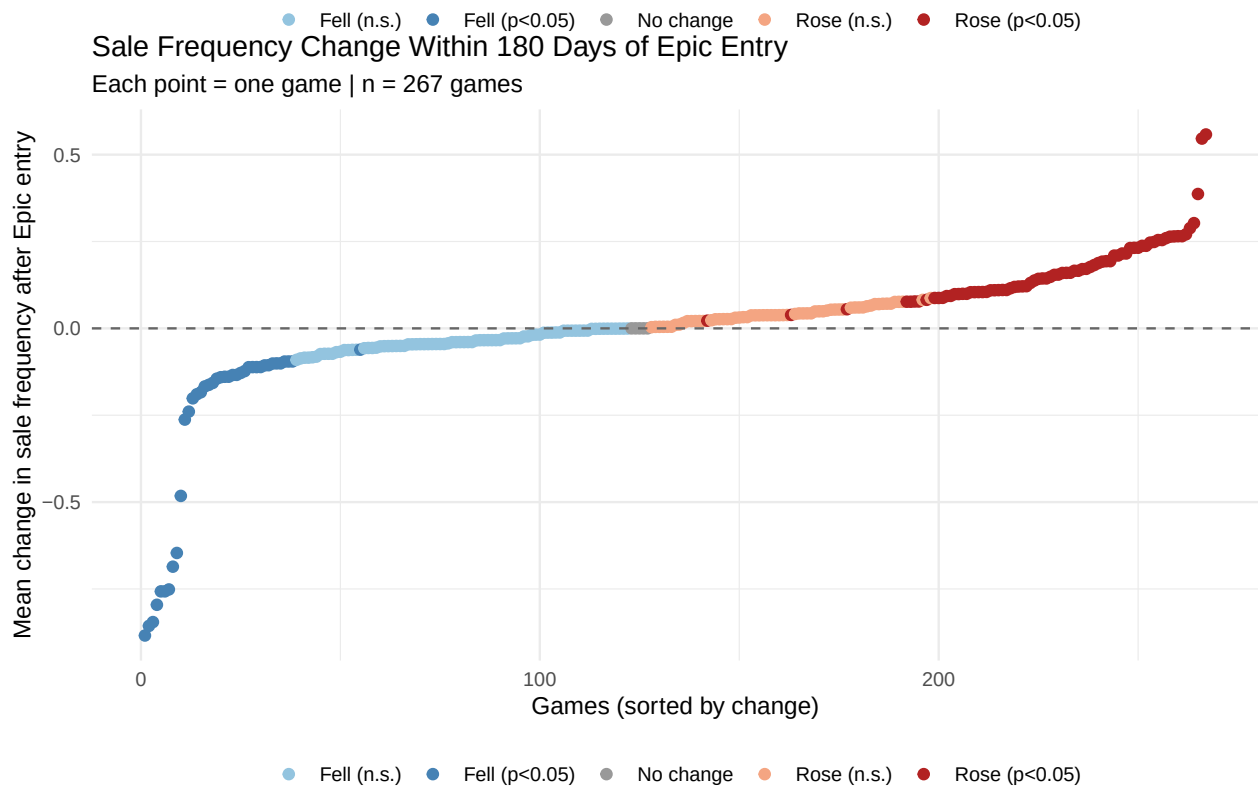
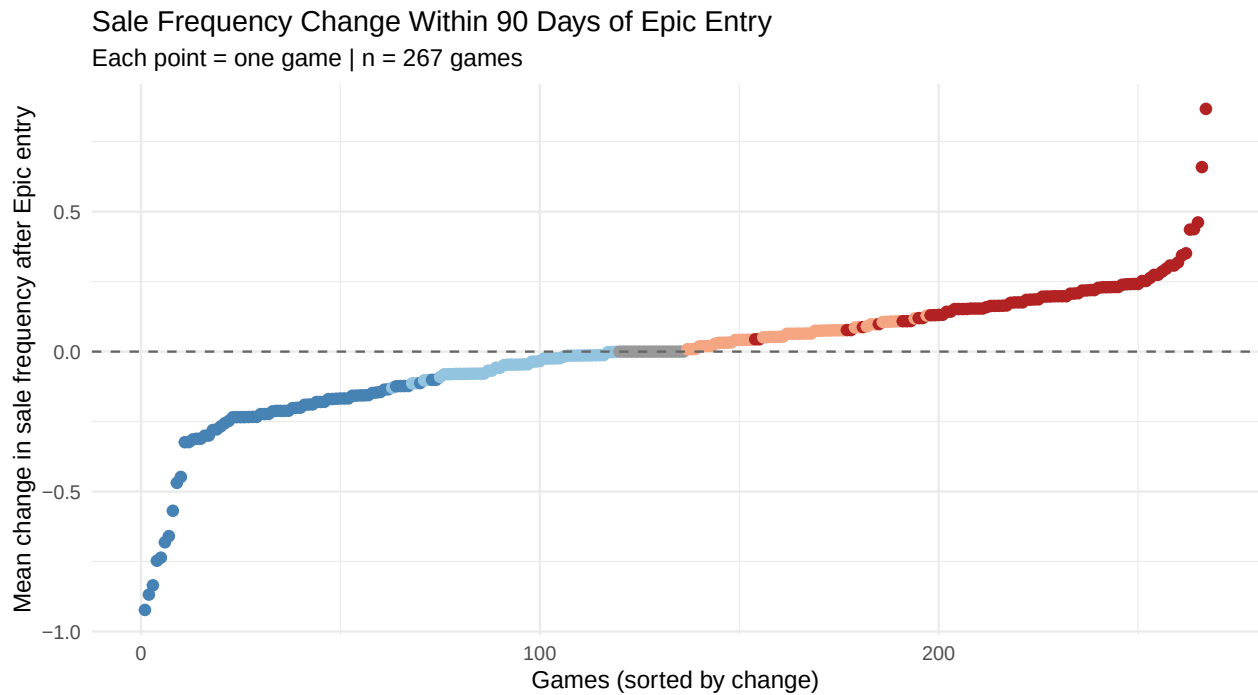
Each point = one game | n = 267 games



5. Sale-frequency t-test plots (one per window)

```
for (w in windows) {
  print(plot_ttests(ttest_results, "Sale Frequency", w,
    "Mean change in sale frequency after Epic entry"))
}
```





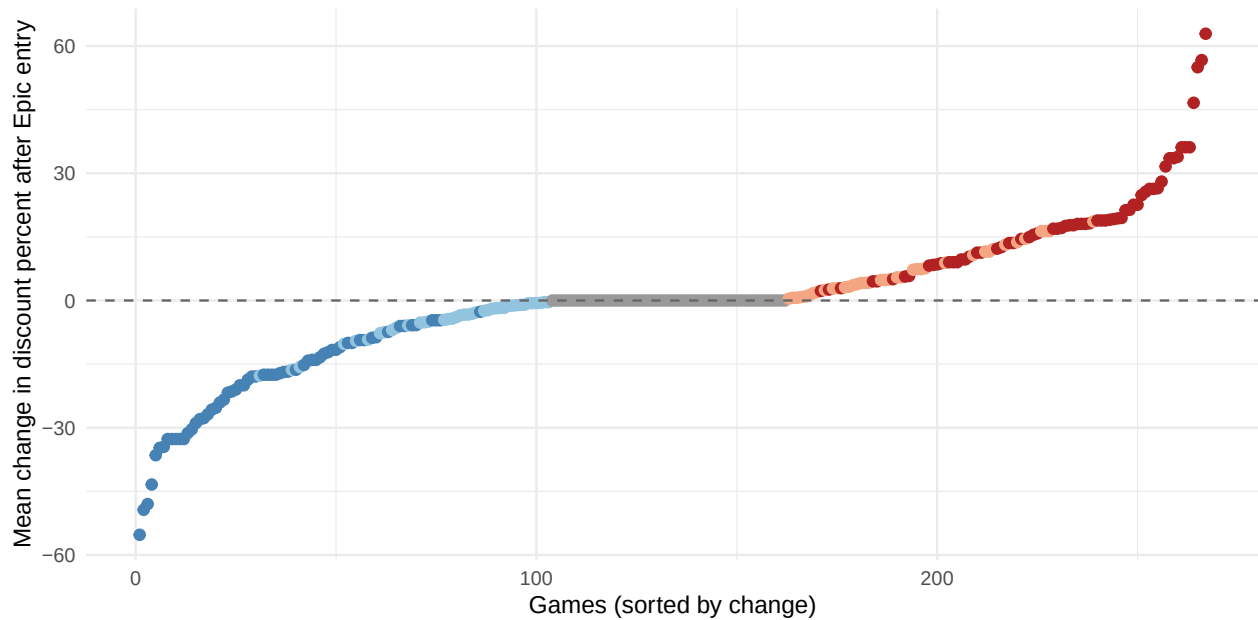
6. Discount-depth t-test plots (one per window)

```
for (w in windows) {
  print(plot_ttests(ttest_results, "Discount Depth", w,
    "Mean change in discount percent after Epic entry"))
}
```

3

Discount Depth Change Within 30 Days of Epic Entry

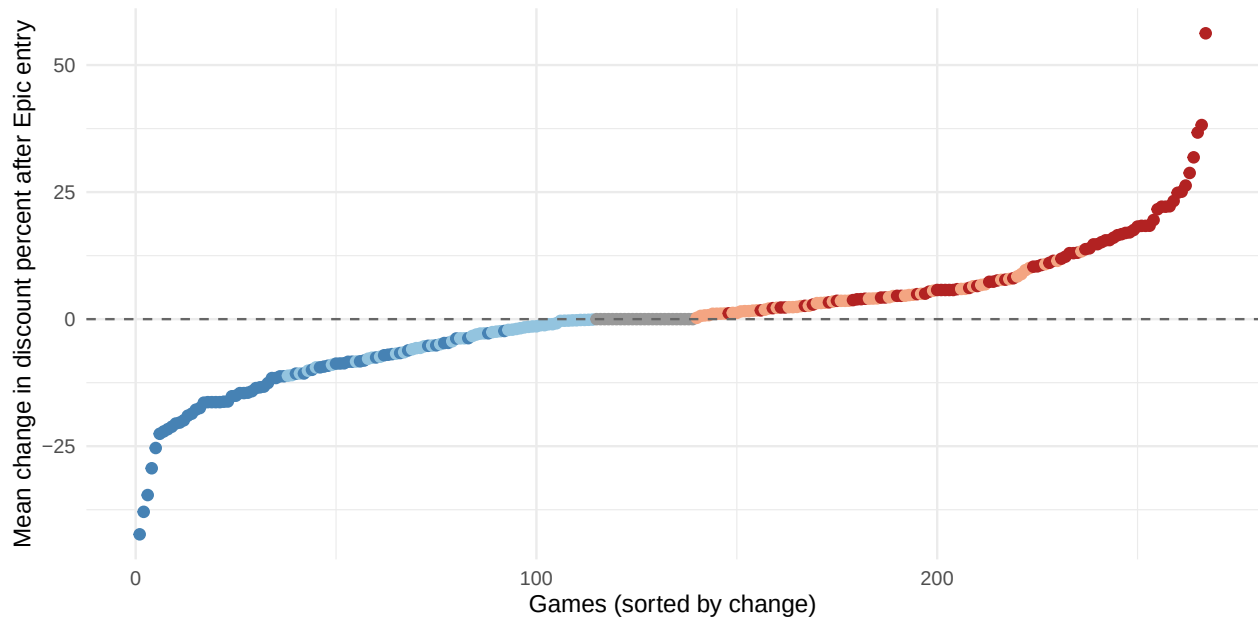
Each point = one game | n = 267 games



Fell (n.s.) Fell (p<0.05) No change Rose (n.s.) Rose (p<0.05)

Discount Depth Change Within 60 Days of Epic Entry

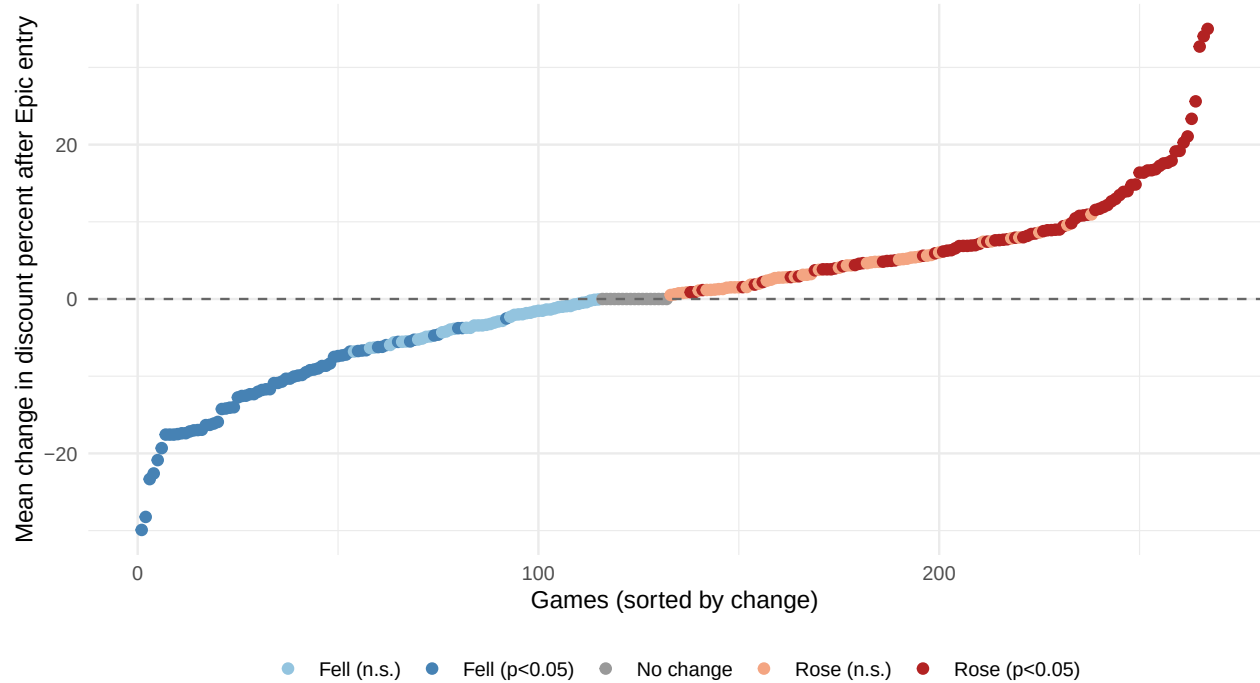
Each point = one game | n = 267 games



Fell (n.s.) Fell (p<0.05) No change Rose (n.s.) Rose (p<0.05)

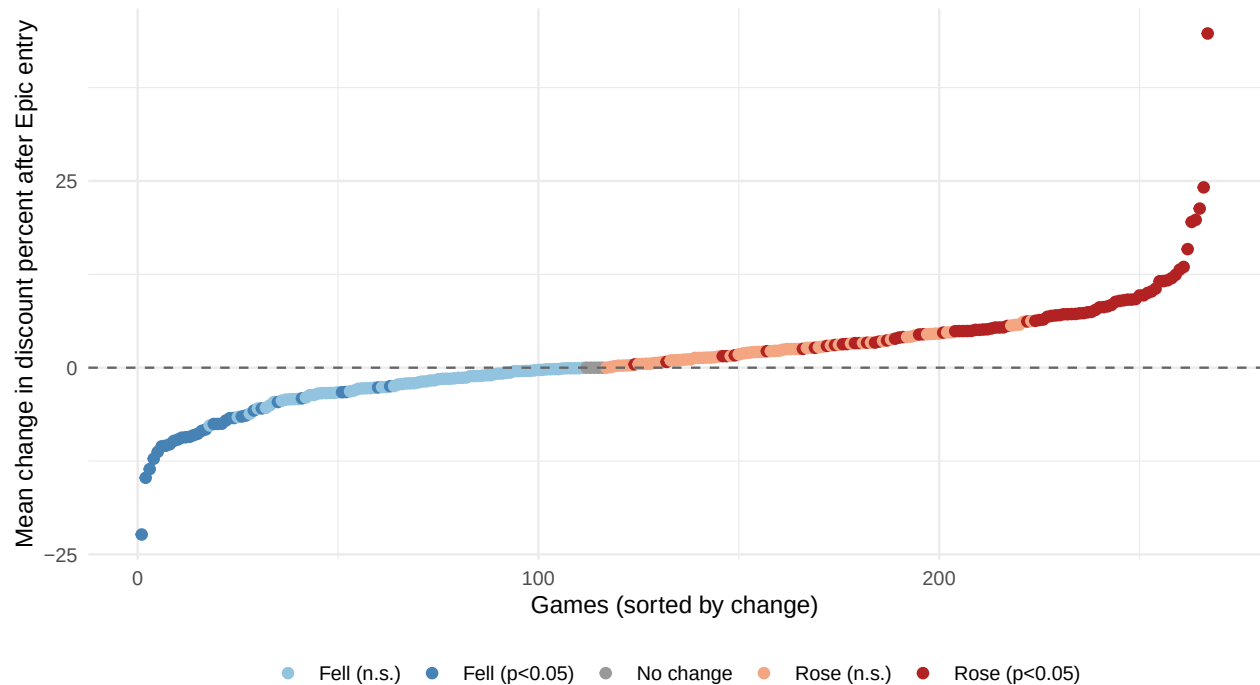
Discount Depth Change Within 90 Days of Epic Entry

Each point = one game | n = 267 games



Discount Depth Change Within 180 Days of Epic Entry

Each point = one game | n = 267 games



7. Summary counts across all outcomes and windows

```
ttest_results %>%
  group_by(outcome, window) %>%
  summarise(
```

```

n_games = n(),
n_rose = sum(direction_up),
n_fell = sum(!direction_up),
n_sig_rose = sum(direction_up & significant),
n_sig_fell = sum(!direction_up & significant),
.groups = "drop"
) %>%
arrange(outcome, window)

## # A tibble: 12 x 7
##   outcome      window n_games n_rose n_fell n_sig_rose n_sig_fell
##   <chr>      <dbl>   <int> <int> <int>   <int>   <int>
## 1 Discount Depth    30     267   105   162     66     63
## 2 Discount Depth    60     267   128   139     74     64
## 3 Discount Depth    90     267   135   132     83     67
## 4 Discount Depth   180     267   151   116     83     33
## 5 Price           30     267   105   162     65     68
## 6 Price           60     267   114   153     67     72
## 7 Price           90     267   117   150     72     80
## 8 Price          180     267   115   152     47     85
## 9 Sale Frequency   30     267   107   160     65     63
## 10 Sale Frequency  60     267   123   144     73     62
## 11 Sale Frequency  90     267   131   136     81     69
## 12 Sale Frequency 180     267   140   127     77     39

ttest_summary <- ttest_results %>%
  group_by(outcome, window) %>%
  summarise(
    n_rose = sum(direction_up),
    n_fell = sum(!direction_up),
    n_sig_rose = sum(direction_up & significant),
    n_sig_fell = sum(!direction_up & significant),
    .groups = "drop"
  ) %>%
  mutate(outcome = factor(outcome,
    levels = c("Price", "Sale Frequency", "Discount Depth"))) %>%
  arrange(outcome, window)

ttest_ft <- ttest_summary %>%
  flextable() %>%
  set_header_labels(
    outcome = "Outcome",
    window = "Window (days)",
    n_rose = "Rose",
    n_fell = "Fell",
    n_sig_rose = "Rose (p<0.05)",
    n_sig_fell = "Fell (p<0.05)"
  ) %>%
  merge_v(j = "outcome") %>%
  valign(j = "outcome", valign = "top") %>%
  theme_booktabs() %>%
  align(j = 2:6, align = "center", part = "all") %>%
  align(j = "outcome", align = "left", part = "all") %>%
  autofit() %>%

```

```
fix_border_issues()
```

```
ttest_ft
```

```
## Warning: fonts used in `flextable` are ignored because the `pdflatex` engine is
## used and not `xelatex` or `lualatex`. You can avoid this warning by using
## `set_flextable_defaults(fonts_ignore=TRUE)` or use a compatible engine (e.g.
## `pdf-engine: xelatex`).
```

Outcome	Window (days)	Rose	Fell	Rose (p<0.05)	Fell (p<0.05)
Price	30	105	162	65	68
	60	114	153	67	72
	90	117	150	72	80
	180	115	152	47	85
Sale Frequency	30	107	160	65	63
	60	123	144	73	62
	90	131	136	81	69
	180	140	127	77	39
Discount Depth	30	105	162	66	63
	60	128	139	74	64
	90	135	132	83	67
	180	151	116	83	33

```
save_as_docx(ttest_ft, path = "Output/tab_ttest_summary.docx")
```

```
ttest_summary_df <- ttest_summary %>%
  mutate(outcome = as.character(outcome)) %>%
  as.data.frame()

stargazer(
  ttest_summary_df,
  summary = FALSE,
  rownames = FALSE,
  type = "latex", # "html" to open in Word, "latex" for a PDF build
  digits = 0,
  title = "T-Test Outcomes",
  covariate.labels = c("Outcome", " Window (days) ",
                       " Rose ", " Fell ", " Rose (p<0.05) ", " Fell (p<0.05) "),
  out = "Output/tab_ttest_summary.html"
)
```

```
##
```

```
## % Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac@sp.i.cas.cz
```

```
## % Date and time: Wed, Sep 23, 2026 - 10:22:15
```

```
## \begin{table}[!htbp] \centering
```

```
## \caption{T-Test Outcomes}
```

```
## \label{}
```

```

## \begin{tabular}{@{\extracolsep{5pt}} cccccc}
## \[-1.8ex]\hline
## \hline \[-1.8ex]
## Outcome & Window (days) & Rose & Fell & Rose (p<0.05) & Fell (p<0.05) & \\\
## \hline \[-1.8ex]
## Price & $30$ & $105$ & $162$ & $65$ & $68$ & \\\
## Price & $60$ & $114$ & $153$ & $67$ & $72$ & \\\
## Price & $90$ & $117$ & $150$ & $72$ & $80$ & \\\
## Price & $180$ & $115$ & $152$ & $47$ & $85$ & \\\
## Sale Frequency & $30$ & $107$ & $160$ & $65$ & $63$ & \\\
## Sale Frequency & $60$ & $123$ & $144$ & $73$ & $62$ & \\\
## Sale Frequency & $90$ & $131$ & $136$ & $81$ & $69$ & \\\
## Sale Frequency & $180$ & $140$ & $127$ & $77$ & $39$ & \\\
## Discount Depth & $30$ & $105$ & $162$ & $66$ & $63$ & \\\
## Discount Depth & $60$ & $128$ & $139$ & $74$ & $64$ & \\\
## Discount Depth & $90$ & $135$ & $132$ & $83$ & $67$ & \\\
## Discount Depth & $180$ & $151$ & $116$ & $83$ & $33$ & \\\
## \hline \[-1.8ex]
## \end{tabular}
## \end{table}

```

```

#For Word: type = "html", out = "Output/tab_ttest_summary.html"

```